D06 · Logistics Technology & Innovation

02 · Warehouse automation: robotics, AGVs & goods-to-person systems

Supply Chain Reference Architecture v2.2 · mrsupplychain.ai

1 WHAT IT IS

Warehouse automation replaces or assists manual handling with equipment and robotics — conveyors and sorters, AGVs and AMRs, goods-to-person systems, AS/RS — chosen according to the operation's order profile, volume and labor economics.

2 WHY IT MATTERS

Automation can multiply productivity and accuracy, but it also fixes the operation's shape in steel and software. **It performs best where volume and process are stable, and poorly where they are not (Azadeh, de Koster and Roy, 2019 — Transportation Science, review of robotized warehouses)**. The decision belongs to a business case with full total cost of ownership, not to a vendor demonstration.

3 HOW IT IS DONE

1. **Profile the operation.** Order lines, units per line, SKU count, seasonality and growth determine which technologies fit.
2. **Stabilize the process first.** Automating a broken process locks in its problems.
3. **Build full TCO and scenarios.** Capex, integration, maintenance, software and specialized talent; test volume downside.
4. **Plan the people transition.** Fewer operators, more technicians and analysts — redesign roles and training before go-live.

IN PRACTICE

Build the business case with full TCO

Include maintenance, software, integration and specialized labor. Optimistic cases that ignore these items create future credibility problems.

Match technology to order and volume profile

The best technology for a high-volume unit-load operation is rarely the best technology for a high-SKU, low-lines e-commerce operation.

Plan the talent transition explicitly

Automation reduces operational headcount but increases the need for technical and analytical skills. The organization must be redesigned, not just the building.

ILLUSTRATIVE CASE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case — E-commerce fulfillment center, 18,000 active SKUs, high labor turnover

Metric	Before (manual)	After (Goods-to-Person)
Picking productivity	68 lines/hour	240 lines/hour
Picking accuracy	98.4%	99.97%
Picking FTEs	180	45
Peak capacity constraint	Labor availability	Largely removed

Result: Capex $4.8M, payback ~2.8 years on labor and error reduction alone. The strategic benefit of reliable peak capacity was judged even more valuable than the pure financial return.

HOW IT IS MEASURED

Picking Productivity (lines or units per hour)

Output per labor hour or per robot hour

Benchmark: Manual 60–120 lines/h · GTP systems commonly 200–400+

Picking Accuracy %

(Correct lines / Total lines picked) × 100

Benchmark: >99.9% expected from mature automated systems

WHAT YOU WOULD USE

Warehouse Automation

AutoStore / Exotec / Swisslog / Knapp (GTP & AS/RS) — Leading goods-to-person and automated storage solutions.

Locus / Geek+ / other collaborative AMR vendors (AMR collaborative picking) — Lower-capex collaborative robot options that often deliver faster payback in mid-sized operations.

4 SO WHAT

Fit the technology to the order profile and the business case. Stabilize the process, count the full TCO, test downside scenarios and redesign the organization along with the building.

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